

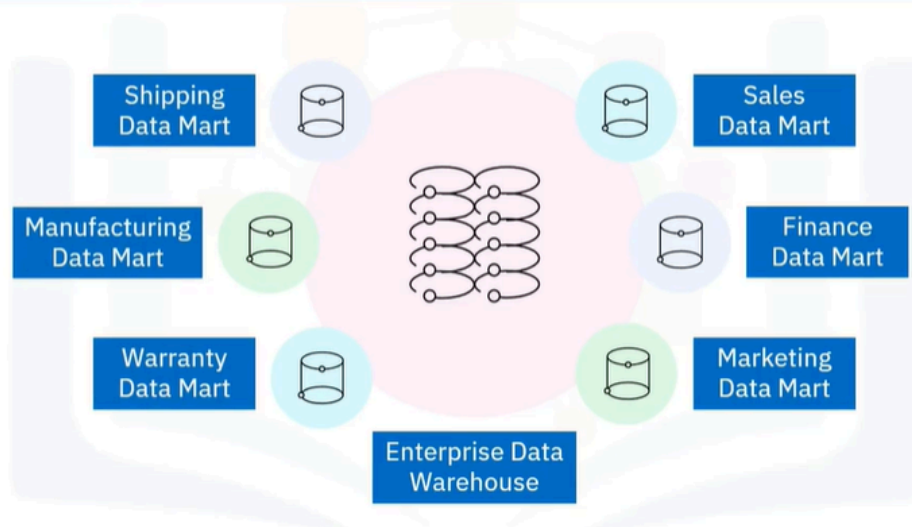
Data Marts

Sure! Let's talk about data marts in simple terms.

A data mart is like a small, specialized store of information within a larger data warehouse. Imagine a big supermarket (the data warehouse) that has everything you could ever need. Now, think of a data mart as a small section of that supermarket that focuses only on a specific type of product, like snacks or beverages. This makes it easier for people, like the sales or marketing teams, to find exactly what they need without wandering through the entire store. Data marts are designed to help businesses make quick decisions by providing relevant data without the hassle of searching through a lot of unrelated information.

For example, if the marketing team wants to analyze customer behavior, they can go straight to their data mart, which has all the necessary data neatly organized for them. This saves time and helps them focus on what matters most for their work.

What is a data mart?



- A data mart is an isolated part of the larger enterprise data warehouse that is specifically built to serve a particular business function, purpose, or community of users. For example, the sales and finance departments in a company may have access to dedicated data marts that supply the data required for their quarterly sales reports and projections. The marketing team may use data marts to analyze customer behavior data, and the shipping, manufacturing and warranty departments may have their own data marts.

What are data marts used for?

Data marts are used to:

- Provide support for tactical decision-making
- Help end users focus only on relevant data
- Save time otherwise spent searching the data warehouse for answers

- So, what are data marts used for? Data marts are designed to provide specific support for making tactical decisions. As such, data marts are focused only on the most relevant data, which saves end users the time and effort that would otherwise be spent searching the data warehouse for insights.

Data mart structure

Typical structure of a data mart:

- Relational database
- Star or snowflake schema
- Central fact table of business metrics
- Surrounded by associated dimension tables

- The typical structure of a data mart is as follows: It is a relational database with a star, or more often a snowflake schema, which means it contains a central fact table consisting of the business metrics relevant to a business process, which is surrounded by a related hierarchy of dimension tables that provide context for the facts.

Data repository comparisons

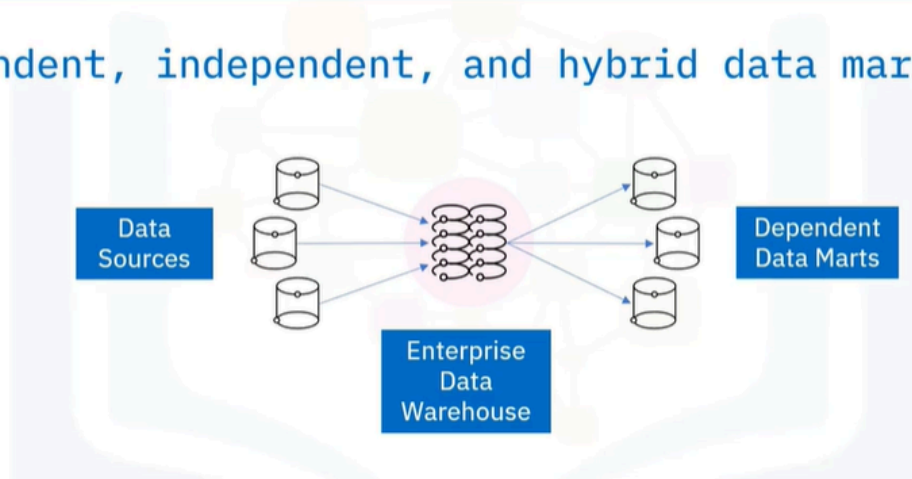
Data Marts	Databases
OLAP systems – read intensive	OLTP systems – write intensive
Use Txn DBs or warehouses as data sources	Use operational applications as sources of data
Contain clean, validated analytical data	Contain raw, unprocessed transactional data
Accumulate history for trend analysis	May not always store history

Data Marts	Data Warehouses
Small data warehouses with tactical scope	Large repositories with broad, strategic scope
Lean and fast	Large and slow

- Let's look at some typical differences between three types of data repositories: data marts, transactional databases, and data warehouses, starting with data marts and databases. Both data marts and data warehouses are online analytical processing (or OLAP) systems that are optimized for read-intensive queries and operations, whereas transactional databases are online transaction processing (or OLTP) systems that are optimized for write-intensive queries and applications.
- Data marts use transactional databases or data warehouses as data sources, while in transactional databases, operational applications, such as point-of-sales systems, serve as the sources of data. A data mart stores validated, transformed, and cleaned data, while a database will have raw data that has not yet been cleaned. Data marts accumulate historical data that can be used for trend analysis, while transactional databases may not always store older data. A data mart is much like a data warehouse, except it has a smaller, tactical scope.
- Data warehouses broadly support the strategic requirements of the enterprise. Data marts are lean and fast compared to data warehouses, which can be very large, and hence, can be slower.

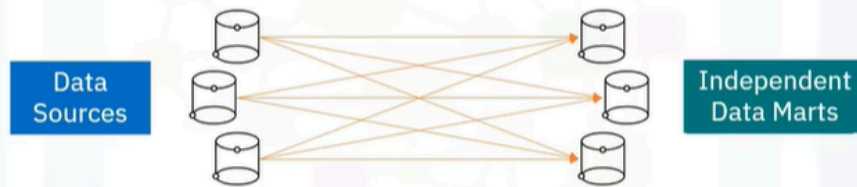
Types of data marts

Dependent, independent, and hybrid data marts



Types of data marts

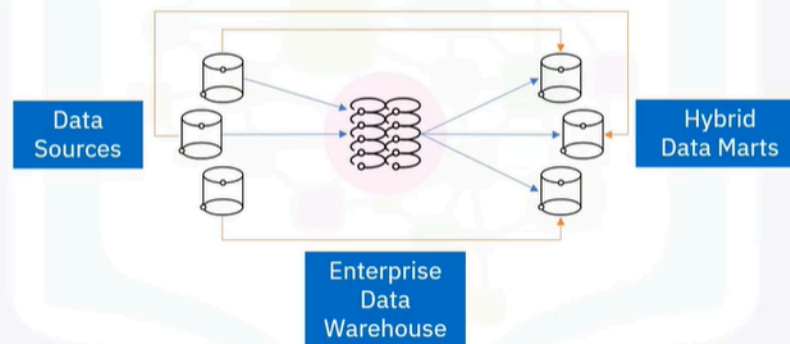
Dependent, independent, and hybrid data marts



- There are three basic types of data marts—dependent, independent, and hybrid. The difference between these three kinds of data marts depends on their relationship with the data warehouse and the sources used for supplying each of them with data.
- Dependent data marts draw data from the enterprise data warehouse, while independent data marts bypass the data warehouse and are created directly from sources, which may include internal operational systems or external data from vendors or other sources outside the enterprise.

Types of data marts

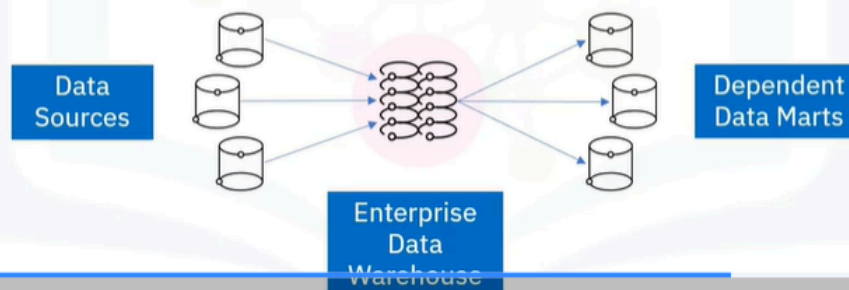
Hybrid data marts



Dependent data marts

Dependent data marts:

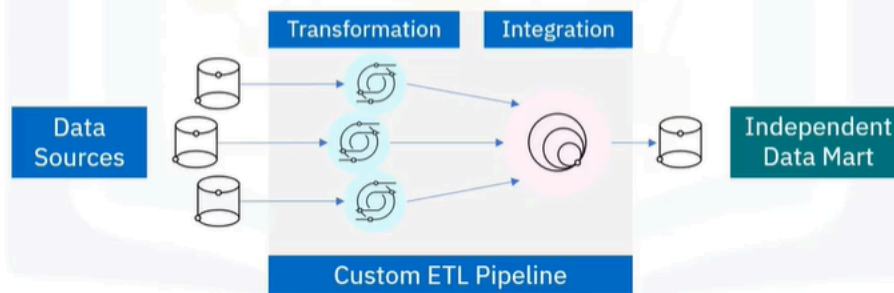
- Inherit security from the EDW
- Use cleaned and transformed data
- Have simpler data pipelines



Independent data marts

Independent data marts:

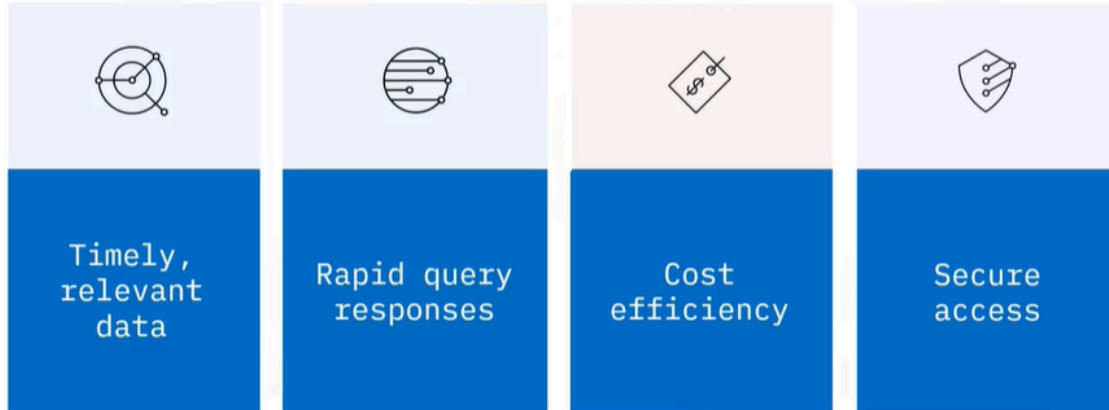
- Require custom ETL data pipelines
- May require additional security measures



- Hybrid data marts only depend partially on the enterprise data warehouse. They combine inputs from data warehouses with data from operational systems and other systems external to the warehouse.
- Dependent data marts offer analytical capabilities within a restricted area of the enterprise data warehouse. Thus, they inherit the security that comes with the enterprise data warehouse. And since dependent data marts pull data directly from the data warehouse, where data has already been cleaned and transformed, they tend to have simpler data pipelines than independent data marts.
- Independent data marts differ from dependent data marts because they require custom extract, transform and load data pipelines to carry out the transformation and integration processes on the source data since it is coming directly from operational systems and external sources, and independent data marts may also require separate security measures.

Data mart purpose

The purpose of a data mart is to provide:



- Whatever the type of data mart you may have, its purpose is to: Provide end-users with relevant data when they need it. Accelerate business processes by providing efficient query response times. Provide a cost-efficient method for informing data-driven decisions, and Ensure secure access and control over your data.

Summary

In this video, you learned that a data mart:

- Is an isolated part of the larger EDW, built to serve a business function, purpose, or user community
- Is designed to provide specific, timely, and rapid support for making tactical decisions
- Typically has a star or snowflake schema
- Accumulates clean and validated historical data
- Can be categorized in relation to the EDW: dependent, independent, or a hybrid of the two

